

**SASTRA DEEMED UNIVERSITY**  
(A University under section 3 of the UGC Act, 1956)

19.

**End Semester Examinations**

**Nov 2025**

Course Code: **CIV101**

Course: **BASIC CIVIL ENGINEERING**

QP No. : **S1726-1**

Duration: **3 hours**

Max. Marks: **100**

**PART – A**

**Answer all the questions**

**10 x 2 = 20 Marks**

1. Identify the branch which deals with the design of foundations and the design of structural members of a multistoried building project.
2. A plot of ground has an area of 600 sq.m. It has built up area of Ground floor 250 sq.m, first floor 200 sq.m and second floor 150 sq.m. Find the FSI and check whether it is within permissible limits.
3. List out any four tests, which will be practiced to ensure the quality of brick.
4. Contrast the different types of loads, acting on the structure constructed for the public utility.
5. Brief the following terms:
  - a) King closer of the brick masonry.
  - b) Tread of a step.
6. Find the super elevation for a horizontal curve, having a radius of 400m. Assume the average speed of the vehicle as 60 Km/hour.

7. Convert the following the whole circle bearing to the reduced bearing.
  - a)  $160^{\circ}30'$
  - b)  $320^{\circ}30'$
8. Outline the importance of foresight and change point practiced in the levelling survey work.
9. Distinguish a storage dam and a diversion dam.
10. State the necessity of irrigation.

### **PART - B**

**Answer all the questions**

**4 x 15 = 60 Marks**

11. Briefly describe the functions performed by a civil engineer before commencement, during progress and after completion in a civil engineering project.

**(OR)**

12. a) Describe briefly any five types of concrete practised in construction works. (10)
- b) State the properties and uses of high-carbon steel. (5)
13. a) With neat sketches, explain the types of Ashlar masonry wall construction works. (10)
- b) Distinguish between a one-way slab and a two-way slab adopted in residential building construction. (5)

**(OR)**

14. Explain with neat sketches, any three types of doors and any two types of windows.
15. The following perpendicular offsets were taken at 10m intervals from a survey line to the irregular boundary line:

3.25, 5.6, 4.2, 6.65, 8.75, 6.2, 3.25, 4.2 and 5.65

Calculate the area enclosed between the survey line to irregular boundary line and first & last offsets by the application of

- a) Trapezoidal rule. (7)
- b) Simpson's rule. (8)

**(OR)**

16. a) The following staff readings were observed successively at a level. 0.875, 1.235, 2.310, 1.385, 2.930, 3.125, 4.125, 0.120, 1.875, 2.030, 2.765. The first reading was taken with a staff held on a benchmark of elevation 132.135 m. The instrument had been moved after the second, fifth and eighth readings. Enter the above readings in a page of a level book and calculate the reduced levels by the HCL method. (10)
- b) Briefly, the basic characteristics of a contour map. (5)
17. Explain the following with neat sketches.
  - a) Check dams. (5)
  - b) Filter pit of a Rainwater harvesting. (5)
  - c) Recharge well. (5)

**(OR)**

18. Describe with a neat sketch any three types of rigid dams and any two types of non-rigid dams.

### **PART - C**

**Answer the following**

**1 x 20 = 20 Marks**

19. a) Describe briefly the various types of deep foundations with neat sketches. (10)
- b) Explain in detail the qualities of good stone. (10)

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